

Answers

2. The rational canonical form for A is

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 5 \end{pmatrix}.$$

3. $\mathbb{Z}_{10} \oplus \mathbb{Z}_{30} \cong \mathbb{Z}_5 \oplus \mathbb{Z}_6 \oplus \mathbb{Z}_{10}$ only.

4. Suppose first that I is simple. From Theorem 8.5, we can multiply by suitable matrices from the left to perform row operations on a non-zero element A of I . So if A has two linearly independent rows, appealing to Theorem 8.5(ii), we can find a matrix B in I which has only one non-zero row. Then clearly RB is a non-zero left submodule of I , and since I is simple, we must have $I = RB$. Thus every non-zero matrix in I is of rank 1.

Conversely, of all non-zero matrices of I are of rank 1, we claim that all rows of every matrix in I are pairwise linearly dependent. Indeed, if there are two matrices A and B in I which have two rows that are linearly independent (that is, one row of A together with one row of B), then using Theorem 8.5(ii) again, we find matrices A', B' in I (since it is a left R -module) which have all but one row non-zero, and the non-zero rows are exactly those two linearly independent rows. Swapping the rows of the second matrix B' if necessary, upon adding these two matrices, i.e. $A' + B'$, we see that we get a matrix of rank 2 in I , a contradiction.

5. Write e_1, e_2 and f_1, f_2 for the respective bases for $\mathbb{R}^2$ for which the corresponding linear maps are given by the matrices A and B respectively.

One can check that $e_i \otimes f_j$ yield a basis for $\mathbb{R}^4 \cong \mathbb{R}^2 \otimes_{\mathbb{R}} \mathbb{R}^2$. We have that

$$\begin{aligned} (A \otimes_{\mathbb{R}} B)(e_1 \otimes f_1) &= (e_1 + 3e_2) \otimes (5f_1 + 7f_2) \\ &= 5(e_1 \otimes f_1) + 7(e_1 \otimes f_2) + 15(e_2 \otimes f_1) + 21(e_2 \otimes f_2), \\ (A \otimes_{\mathbb{R}} B)(e_1 \otimes f_2) &= (e_1 + 3e_2) \otimes (6f_1 + af_2) \\ &= 6(e_1 \otimes f_1) + a(e_1 \otimes f_2) + 18(e_2 \otimes f_1) + 3a(e_2 \otimes f_2), \\ (A \otimes_{\mathbb{R}} B)(e_2 \otimes f_1) &= (2e_1 + 4e_2) \otimes (5f_1 + 7f_2) \\ &= 10(e_1 \otimes f_1) + 14(e_1 \otimes f_2) + 20(e_2 \otimes f_1) + 28(e_2 \otimes f_2), \\ (A \otimes_{\mathbb{R}} B)(e_2 \otimes f_2) &= (2e_1 + 4e_2) \otimes (6f_1 + af_2) \\ &= 12(e_1 \otimes f_1) + 2a(e_1 \otimes f_2) + 24(e_2 \otimes f_1) + 4a(e_2 \otimes f_2). \end{aligned}$$

$$\text{Hence } C = \begin{pmatrix} 5 & 6 & 10 & 12 \\ 7 & a & 14 & 2a \\ 15 & 18 & 20 & 24 \\ 21 & 3a & 28 & 4a \end{pmatrix} \text{ and } \det C \neq 0 \text{ for } a \neq 42/5.$$